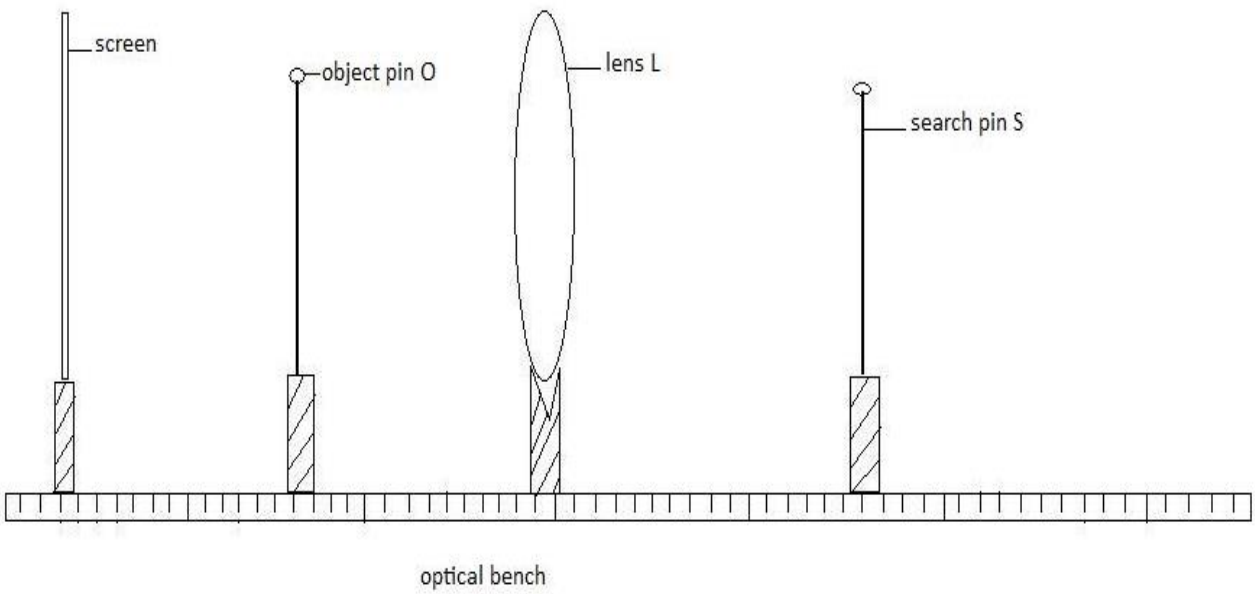
AIM

The aim is to determine the focal length of a converging lens.

APPARATUS

- Convex lens (1)
- Optical pins (2)
- Metre rule (1)
- White screen (1)

DIAGRAM



THEORY

When a group of rays parallel to the axis pass through the focal point after being converged by the lens, the distance between the focal point and the lens is referred to as the focal distance or length.

Magnification $m = \frac{image\ distance}{object\ distance}$

From this experiment, $l - d$ is the image distance and d is the object distance

Also, $m = \frac{v}{f} - 1$

A graph of magnification against image distance yields a straight line with a negative intercept.

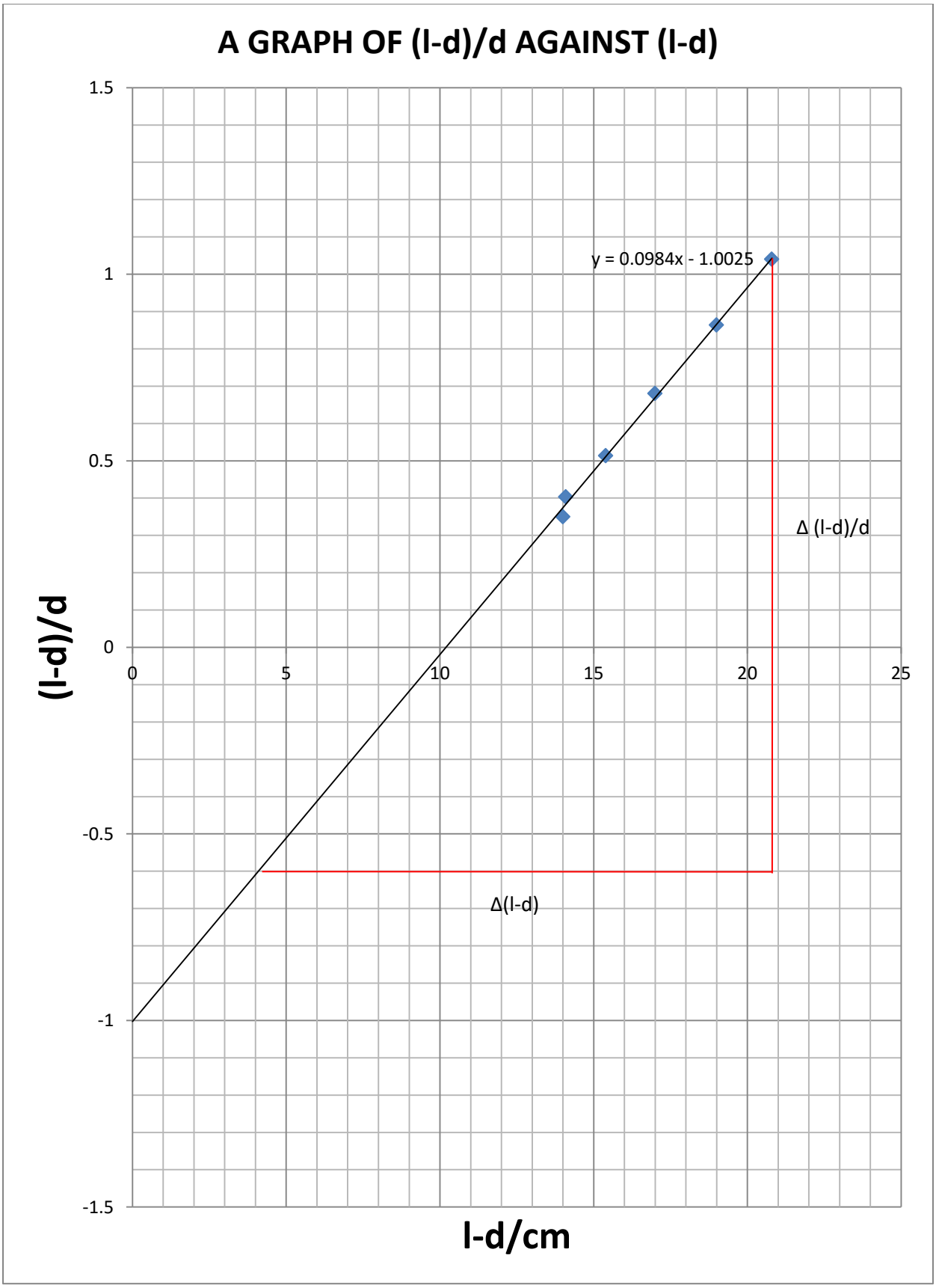
PROCEDURE

A screen, object pin O, the lens L, and a search pin S were placed on an optical bench as in the diagram above. The pins were placed in such a way that their tips were lying approximately along the axis of the lens.

The object pin O was positioned at a distance from the lens. An eye was placed behind the search pin S. the search pin was moved towards or away from the lens until it was in no parallax position with the image of the object pin. The distance between the search pin and the object pin was measured as l. l-d was calculated. The experiment was repeate for other values of the object distance d and in each case l-d, $\frac{l-d}{d}$ was calculated. The results were tabulated and a graph of $\frac{l-d}{d}$ as ordinate and $l - d$ as abscissa was plotted. The slope was calculated and the intercept determined from the graph. The reciprocal of the slope was also found and its significance indicated.

TABLE OF RESULT

d/cm	l/cm	$l - d/cm$	$\frac{l-d}{d}/cm$
20.0	40.8	20.8	1.0400
22.0	41.0	19.0	0.8636
25.0	42.0	17.0	0.6800
30.0	45.4	15.4	0.5133
35.0	49.1	14.1	0.4029
40.0	54.0	14.0	0.3500


A GRAPH OF (l-d)/d AGAINST (l-d)
1.5
1
0.5
0
-0.5
-1
-1.5
0
5
10
15
20
25
y = 0.0984x - 1.0025
Δ (l-d)/d
Δ(l-d)
(l-d)/d
l-d/cm

CALCULATION

$$\text{slope(m)} = \frac{\Delta\frac{\text{l}-\text{d}}{\text{d}}}{\Delta(\text{l}-\text{d})} = \frac{1.04-(-0.6)}{20.8-4.0} = \frac{1.64}{16.8} = 0.0976cm^{-1}$$

Using $y = mx + c$ where m=0.0976cm^{-1}, y=1.04 and x=20.8

$$=> \quad 1.04 = (0.0976cm^{-1})(20.8cm) + c$$

$$c = 1.04 - 2.03 = -0.99 \approx -1$$

The reciprocal of the slope (m) = $\frac{1}{0.0976cm^{-1}} = 10.2459cm \approx 10.2cm$

DISCUSSION

Using the lens equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, one was able to derive the equation $m = \frac{v}{f} - 1$.The graph to be produced by the experiment using that equation had to result in a straight line passing through the intercept of -1. As shown above, this was achieved. The straight line of the graph resulted in a slope of 0.0976cm^{-1} which represented $\frac{1}{f}$ from the equation given. So the reciprocal of the slope will give us the focal length (f) of the lens used and this was 10.2cm approximately.

The intercept was also determined which was -0.99. This is approximately -1 which tells us that the equation is valid since the theoretical intercept corresponds with the experimental intercept.

SOURCES OF ERRORS

1. The search pin as well as the object pin were too loosely attached to the optical bench

PRECAUTIONS

1. The eye was placed exactly in front of the search pin and not at an angle above or below the pins.
2. The lens was wiped thoroughly to make it clean and clear when looking through it.
3. A screen was placed behind the set up too ensure a clear view of the images formed.
4. The experiment was set up in a well lighted surrounding to produce clear images.

CONCLUSION

The focal length of the converging lens used in the experiment is 10.2cm.